# Topic: Micro-Operations

**Q. What are Micro-Operations? Discuss the four main types: Register Transfer, Arithmetic, Logic, and Shift Micro-Operations with examples.**

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> 📌 **Definition to Remember**
> A **Micro-Operation** is the most basic, fundamental, atomic operation performed on data stored in CPU registers during one clock pulse. Every high-level machine instruction is broken down into a sequence of these elementary micro-operations.

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## 1. Types of Micro-Operations

There are four primary categories of micro-operations based on their function.

### A. Register Transfer Micro-Operations

Transfers data from one register to another without modifying the actual data.
* **Notation:** `←` indicates transfer.
* **Example:** `R2 ← R1` (Copies data from R1 into R2. R1 is unchanged).
* **Control:** Usually triggered by a control condition: `If (P=1) then R2 ← R1`.

### B. Arithmetic Micro-Operations

Performs basic math on numeric data in registers.
* **Operations:** Add, Subtract, Increment, Decrement, 2's Complement (Negation).
* **Examples:**
* `R3 ← R1 + R2` (Addition)
* `R1 ← R1 + 1` (Increment)
* `R2 ← R1' + 1` (2's Complement/Negate R1, store in R2).

### C. Logic Micro-Operations

Performs bit-by-bit manipulation on non-numeric binary data.
* **Operations:** AND, OR, XOR, NOT.
* **Uses:** Masking bits, clearing registers, inserting bits.
* **Examples:**
* `R1 ← R1 ^ R2` (Bitwise XOR).
* `R1 ← R1 & R2` (Bitwise AND).
* `R1 ← 0` (Can be done by XORing a register with itself: `R1 ← R1 ^ R1`).

### D. Shift Micro-Operations

Moves the binary bits of a register left or right. Used for data alignment and fast multiplication/division.
* **Logical Shift:** Shifts bits, filling the newly empty space with `0`. (`R1 ← shl R1`).
* **Circular Shift (Rotate):** Bits shifted out of one end are fed directly back into the other end.
* **Arithmetic Shift:** Shifts right but preserves the Most Significant Bit (MSB/Sign bit), essential for signed negative numbers.

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⭐ **Must-Write Points (for 10 marks)**

1. **Micro-Operation**: The smallest atomic operation executed in one clock pulse.
2. Four types: **Register Transfer, Arithmetic, Logic, Shift**.
3. **Register Transfer**: Moves data without changing it. Example: `R2 ← R1`.
4. **Arithmetic**: Basic math (Add, Sub, Inc). Example: `R3 ← R1 + R2`.
5. **Logic**: Bitwise operations (AND, OR, XOR) used for masking. Example: `R1 ← R1 ^ R2`.
6. **Shift**: Moves bits left/right for alignment or fast math.
7. Shift types: Logical (fills with 0), Circular (rotates bits), Arithmetic (preserves sign bit).

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**Quick Recall**

`Micro-Ops (1 clock pulse) → 4 Types: Transfer (R2←R1) → Arithmetic (R1+R2, +1) → Logic (AND/XOR, bit masking) → Shift (Logical, Circular, Arithmetic for signed math)`